Session 3 – Gibbs posteriors, hold-out, imbalance and boosting

Goal. By the end of the session you should be able to build voters so that the prior is legitimate, compare self-bounding and hold-out certificates, certify the balanced error, and understand why boosted votes are hard to certify.
To reread before the session. Sections 4, 6 and 7.3 of the notes; notebooks 2 and 4.
Timing (3 hours).

	Part 3.1	50 min
	Part 3.2	35 min
	Part 3.3	50 min
	Part 3.4	45 min

To hand in at the end of the session. Your notebook, with the figure of Part 3.1 and the tables of Parts 3.2 to 3.4.

Part 3.1 – The Gibbs posterior and its temperature (Section 4)

We now build the voters ourselves, so that the prior can be fixed before seeing the data used in the bound.

1. On `wdbc` and `australian`, keep 20% of the data to build the voters: all the decision stumps $x \mapsto s\ \mathrm{sign}(x_j - \theta)$, $s \in \{\pm 1\}$, whose thresholds θ are the quantiles $10, 25, 50, 75, 90\%$ of feature j on this part. Split the remaining data into a sample S (60%) used by the bound and a test sample.

2. With the uniform prior, compute the Gibbs posterior $Q_\lambda \propto e^{-\lambda m \widehat{R}_S(h)}$ for a grid of 60 values of λ between 10^{-3} and $10^{1.5}$, and plot its empirical Gibbs risk, $\mathrm{KL}(Q_\lambda \| P)/m$ and Seeger's bound as functions of λ.

3. Report the best certificate, the corresponding test Gibbs risk and test risk of the vote, and compare with the Occam bound of the stump with the smallest empirical risk. Why is choosing λ on the grid free?

> **Numerical tip.** Compute the Gibbs posterior with a stable softmax, $Q_\lambda = \mathrm{softmax}(\ln P - \lambda m \widehat{R}_S)$, which subtracts the maximum before exponentiating; the naive formula underflows as soon as λm is large.

Part 3.2 – Hold-out certificate versus self-bounding certificate (Section 6.4)

1. *Self-bounding:* learn a forest on the whole training set, minimize its tandem bound, and report its TND certificate. *Hold-out:* learn the same forest on 70% of the training set and compute the kl test-set bound of its (uniform) vote on the remaining 30%. Compare the test risks and the certificates on `spambase`, `satimage`, `pageblocks`, `abalone8`, `wdbc`, `german`.

2. Which approach gives the better vote? the better certificate? Explain the gap between the two certificates.

3. In which situation would you nevertheless recommend the self-bounding approach?

Part 3.3 – Imbalanced datasets and the balanced error (Section 6)

The bounds of this course control the 0-1 risk. On imbalanced datasets a trivial classifier predicting the majority class has a small 0-1 risk.

1. On the imbalanced datasets, compare the certificates of the uniform forest with the risk of the trivial classifier. Are the certificates informative?

2. Obtain a certificate on the *balanced* error $\frac{1}{2}(\mathbb{P}(h(x) \neq y \,|\, y = +1) + \mathbb{P}(h(x) \neq y \,|\, y = -1))$: apply the OOB bounds separately to the training examples of each class (with n_+ and n_- OOB examples) and use a union bound. Implement it and comment.

Part 3.4 – Boosting (Section 7.3)

AdaBoost also outputs a weighted majority vote. To obtain a certificate, the voters must be evaluated on examples independent of them.

1. Split the training set into two halves S_1, S_2. Run `AdaBoostClassifier` with decision stumps (100 rounds) on S_1, and consider the learned stumps as the voters, with the uniform prior.
2. Compute on S_2 the FO and TND certificates of the posterior $\rho_t \propto \alpha_t$ given by AdaBoost (with $\delta/2$ each), and of the posteriors minimizing the two bounds. Compare the test risks with those of AdaBoost trained on the whole training set.
3. Why are the certificates of the AdaBoost posterior loose, although AdaBoost is accurate? *Hint:* compute the Gibbs risk of the stumps on S_2 and the KL of the AdaBoost posterior.